BA 3.2 Capstone Project Documentation

by Tinyiko

General metrics

15,810	2,021	164	8 min 5 sec	15 min 32 sec
characters	words	sentences	reading time	speaking time

Score

99

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Writing Issues

6	2	4
Issues left	Critical	Advanced

Plagiarism

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Writing Issues

4 **Clarity**

4 Paragraph can be improved

2 **Correctness**

2 Wrong or missing prepositions

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

28%

unique words

Rare Words

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37%

rare words

Word Length

Measures average word length

6.4

characters per word

Sentence Length

Measures average sentence length

12.3

words per sentence

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1. **Business objectives**
2. Business Background

Higher education institutions play an important role in supporting student academic achievement, progression, and successful completion of qualifications. However, identifying students who are experiencing academic difficulties can be challenging when assessment results and other academic information are recorded across different systems or reviewed manually. Students may continue to perform poorly on tests, assignments, and practical assessments without fully understanding how their results could affect their ability to pass a module. This creates an opportunity for an intelligent academic monitoring solution that improves access to performance information and supports earlier identification of students who may require assistance. EduAI is proposed as a technology-driven solution for universities and other higher education institutions. The system combines academic performance monitoring with AI-based analytics to show how institutions can identify academic risk, communicate relevant information, and support timely intervention.

2.2 Business Objectives

EduAI's primary business objective is to support student academic success by providing an AI-assisted solution that identifies potential academic difficulties and encourages earlier intervention.

The specific business objectives are to:

- **Identify academic risk:** Analyze student assessment performance and relevant academic indicators to identify students who may be at risk of failing modules.
- **Improve academic awareness:** Provide students with accessible information about their assessment marks, academic averages, and risk indicators.
- **Support informed decision-making:** Demonstrate how AI-generated academic risk information could assist institutions in identifying students who may require additional support.
- **Encourage early intervention:** Provide students with relevant academic-support guidance before final module outcomes are determined.
- **Monitor academic performance:** Use data analysis and time-series techniques to investigate patterns and changes in student academic performance.
- **Demonstrate the application of 4IR:** Apply machine learning, data analytics, deep learning, and natural language processing to a practical problem in higher education.

2.3 Business Success Criteria

The proposed system's success will be measured using both technical and business criteria.

The success of the proposed EduAI system will be evaluated against the following criteria:

- **Academic Risk Identification:** The system should successfully identify and communicate academic risk indicators to help students understand their academic standing.
- **Student Performance Awareness:** Students should be able to access their recorded assessment marks and view their calculated academic averages.
- **Machine Learning Performance:** Train and evaluate Logistic Regression and Random Forest models using relevant historical educational data, measuring performance through accuracy, precision, recall, and F1-score.
- **Academic Support:** The chatbot should provide clear, understandable, and relevant responses to students' academic support questions.

- **Performance Monitoring:** The system should analyze trends in student academic performance to identify improvements or declines over time.
- **System Usability:** Students should be able to navigate the application and complete essential tasks without major difficulties.
- **Business Feasibility:** The project should demonstrate that the proposed solution is technically feasible while identifying the resources, constraints, and risks associated with its implementation in higher education institutions.

These criteria will assess the EduAI prototype's functionality, technical performance, and feasibility. Improvements in actual university pass rates would require further evaluation following institutional implementation.

2.4 Functional Requirements

The proposed System Should:

- **User Authentication:** The system shall allow authorized users to log in using valid credentials.
- **Student Academic Records:** The system shall capture, store, and display students' assessment marks, including tests, assignments, and practical assessments.
- **Academic Average Calculation:** The system shall calculate students' academic averages using their recorded assessment marks.
- **Academic Performance Monitoring**: The system shall identify students whose academic averages fall below the configured 50% performance threshold.
- **Machine Learning:** The AI component shall use trained classification models to analyze relevant historical academic data and identify patterns associated with academic risk.
- **Risk Communication:** The system shall present academic risk information in a clear and understandable format.
- **Student Support Chatbot:** The system should provide a chatbot capable of responding to supported academic-support questions.
- **Performance Trend Analysis:** The analytical component shall demonstrate changes in student academic performance or engagement over time.
- **Data Security:** The system shall restrict access to student academic information according to user permissions.
- **Academic Intervention Support:** The proposed institutional solution shall support academic staff in identifying students who may benefit from additional academic assistance.

2.5 Non-Functional Requirements

The system should:

- **Usability:** The system must provide a simple, understandable, and user-friendly interface.
- **Performance:** The application shall respond to normal user interactions within an acceptable time during demonstrations.
- **Security:** Student academic records must be protected against unauthorized access.
- **Data Integrity:** The system must validate assessment information to reduce incorrect calculations and inconsistent records.
- **Reliability:** The prototype must perform its main functions consistently during testing and demonstration.
- **Maintainability:** The application must be structured to allow future updates and improvements.
- **Understandability:** Risk information and academic guidance must be presented in language accessible to non-technical users.

2.6 Constraints

The development of the system may be affected by the following constraints:

- **Data availability:** Access to real student data may be limited because of privacy and institutional restrictions.
- **Data quality:** Missing, inaccurate, or inconsistent information may reduce AI model performance.
- **Time:** The project must be developed within the academic project deadline.
- **Resources:** The group may have limited computing resources, funding, and access to specialized AI infrastructure.
- **Technical expertise:** The complexity of Machine Learning, Deep Learning, NLP, and Time Series Analysis may require additional research and learning.
- **Privacy:** Student information must be anonymized or protected to prevent the exposure of personal information.
- **Prototype limitations**: The student project will demonstrate a prototype rather than a fully deployed university-wide production system.

2.7Risks

The following risks have been identified for the proposed EduAI solution:

Poor Data Quality

- **Risk:** Missing, incomplete, or inaccurate records may affect academic risk analysis.
- **Solution:** Validate, clean, and preprocess data before analysis.

Incorrect Risk Classification

- **Risk:** Students may receive inaccurate academic risk indicators.
- **Solution:** Evaluate classification models using appropriate metrics and communicate that predictions are estimates.

Data Privacy Breach

- **Risk:** Unauthorized individuals may gain access to student academic information.
- **Solution:** Use de-identified analytical datasets, secure authentication, and appropriate access controls.

AI Bias

- **Risk:** Model predictions may be less reliable for certain student groups.
- **Solution:** Review model performance across relevant groups and investigate significant differences.

Technical Failure

- **Risk:** The application may become unavailable or fail during operation.
- **Solution:** Regularly evaluate the prototype and maintain backup copies.

Limited User Acceptance

- **Risk:** Students or academic staff may be reluctant to use the system.
- **Solution:** Provide an understandable interface and explain how academic risk information supports students.

Limited Generalisability

- **Risk:** Models trained on historical educational data may not perform equally well at other institutions.
- **Solution:** Document dataset limitations and recommend further testing before institutional deployment.

Incomplete System Integration

- **Risk:** Some analytical components may not be fully integrated into the student-facing application.

- **Solution:** Clearly distinguish demonstrated functionality from experimental and proposed features.

2.8 Initial Assessment of Tools and Techniques

- **Python:** Used as the primary programming language for developing AI components and processing educational data.
- **Pandas:** Used to load, clean, organize, and analyze student datasets.
- **NumPy:** Supports numerical calculations and mathematical operations required during data analysis.
- **Scikit-learn:** Provides tools for developing, training, and evaluating machine-learning models.
- **Logistic Regression:** Used as a classification algorithm to identify patterns associated with students' academic risk.
- **Random Forest:** Used as an alternative machine-learning classification algorithm to analyze relationships between student information and academic outcomes.
- **Time-Series Analysis:** Used to examine changes in student academic performance or engagement over time and identify relevant trends.
- **Deep Learning:** Explored as an additional AI technique to identify complex patterns in educational data.
- **Natural Language Processing (NLP):** Enables the chatbot to process and respond to supported questions expressed in natural language.
- **Chatbot:** Provides an interactive communication tool through which students can obtain academic-support information and guidance.
- **Data Visualization:** Used to present model evaluation results, academic performance information, and analytical trends in an understandable format.
- **GitHub:** Used for version control, project collaboration, and tracking development progress.
- **Grammarly:** Used to generate a report.

These tools and techniques support EduAI's development by combining predictive analysis, academic performance monitoring, and student-support functionality. Their application shows how Artificial Intelligence can help higher education institutions identify academic difficulties and support informed decision-making.

1. **Problem Definition**

3.1. What is the problem?

Higher education institutions face challenges in identifying students who are at risk of failing academic modules before their performance deteriorates significantly. Students may obtain poor results in tests, assignments, and practical assessments without fully understanding the consequences for their final module outcomes. Academic staff may also struggle to monitor individual student performance consistently, particularly when managing large student populations. Delayed identification of academic difficulties can reduce opportunities for timely academic support and increase the likelihood of unsuccessful module outcomes.

3.2. Who is affected by the problem?

The problem affects students, lecturers, academic advisors, student-support departments, and university management. Students may not receive support early enough, while academic staff may struggle to identify which students need immediate attention.

3.3 Why is the problem relevant to the theme?

The problem is relevant to the theme “An AI Solution for Industries” because AI can be applied in education to analyze student data, identify patterns, and predict which students may be at risk. The proposed solution uses 4IR technologies to address a real operational and academic challenge.

1. **How will AI solve the problem?**

The proposed system will use Machine Learning to analyze historical student data and predict academic risk levels. Time Series Analysis will monitor changes in student performance over time, while Deep Learning may identify more complex patterns. An NLP-based chatbot will let students and staff interact with the system and get clear information about academic risks.

3.5. How will solving the problem benefit the institution?

The AI solution will help the institution identify at-risk students earlier, allowing academic staff to prioritize interventions and provide appropriate support. This can improve decision-making, make student-support resources more targeted, encourage timely academic assistance, and potentially contribute to improved student progression and retention.

1. **Poster**